

**LESSON 6-3 • HOMEWORK PACKET • LEVEL 1 • SHEET 1 OF 2****Writing Repeated Factors as Powers**

Maryland Standard 6.EE.A.1 · Learning target – I can write repeated multiplication as a power and name its base and exponent.

**REMEMBER**

- A **power** shows a repeated factor:  $5 \times 5 \times 5 = 5^3$ .
- The **base** is the repeated factor. The **exponent** tells how many times it is a factor.
- Read  $5^3$  as "5 to the third power" or "5 cubed." Read  $5^2$  as "5 squared."

**WORKED EXAMPLE**

Write  $6 \times 6 \times 6 \times 6 \times 6$  as a power. Name the base and the exponent.

**Step 1** The repeated factor is 6. It is the base.

**Step 2** Count the factors: there are 5. That is the exponent.

**Step 3** Write the base with a raised exponent:  $6^5$

**Answer**  $6^5$ ; base 6, exponent 5

**WORD BANK**

**power** – a short way to write a repeated factor, like  $4^3$

**base** – the number that repeats

**exponent** – the small raised number: how many times the base repeats

**PRACTICE**

*show your work · label units*

1. Write each product as a power.

a.  $4 \times 4 \times 4 =$  \_\_\_\_\_

b.  $9 \times 9 =$  \_\_\_\_\_

c.  $2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 =$  \_\_\_\_\_

d.  $10 \times 10 \times 10 \times 10 =$  \_\_\_\_\_

e.  $1/3 \times 1/3 \times 1/3 =$  \_\_\_\_\_

f.  $0.5 \times 0.5 =$  \_\_\_\_\_

g.  $6 \times 6 \times 6 \times 6 \times 6 =$  \_\_\_\_\_

h.  $2/5 \times 2/5 =$  \_\_\_\_\_

2. Complete the table.

**Step 1** Base = the repeated number.

**Step 2** Exponent = how many times it is a factor.

| Power | Base | Exponent | Expanded form  |
|-------|------|----------|----------------|
| $8^3$ |      |          |                |
|       | 7    | 4        |                |
|       |      |          | $11 \times 11$ |

- 3. Phone Tree.** A coach calls **3** players. Each player calls **3** more, and so on. In round 5, the number of calls is  **$3 \times 3 \times 3 \times 3 \times 3$** . Write this as a **power**.

**Step 1** The repeated factor is the base.

**Step 2** Count the 3s for the exponent.

**Power** \_\_\_\_\_

- 4. Find the Error.** Rafael wrote  $4 \times 4 \times 4 \times 4 \times 4$  as  $5^4$ . Explain his mistake. Then write the correct power.

**Frame** Rafael switched the \_\_\_\_ and the \_\_\_\_\_. The correct power is \_\_\_\_\_.

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**CHECK YOURSELF**

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #\_\_\_\_\_

**AT HOME** Explain to someone at home why  $2^4$  does not mean  $2 \times 4$ .

**LESSON 6-3 • HOMEWORK PACKET • LEVEL 1 • SHEET 2 OF 2****Finding the Value of a Power**

Maryland Standard 6.EE.A.1 • Learning target – I can find the value of a power by multiplying the base the number of times the exponent shows.

**REMEMBER**

- Write the **expanded form**, then multiply one factor at a time.
- $4^3 = 4 \times 4 \times 4 = 64$ , **not**  $4 \times 3 = 12$ .
- Fraction and decimal bases work the same way:  
 $(1/2)^2 = 1/2 \times 1/2 = 1/4$ .

**WORKED EXAMPLE**

Find the value of  $2^5$ .

**Step 1** Expanded form:  $2 \times 2 \times 2 \times 2 \times 2$

**Step 2** Multiply:  $2 \times 2 = 4$ ,  $4 \times 2 = 8$ ,  $8 \times 2 = 16$ ,  $16 \times 2 = 32$

**Answer**  $2^5 = 32$

**WORD BANK**

**evaluate** – find the value

**expanded form** – the power written as multiplication:  $4^3 = 4 \times 4 \times 4$

**squared** – to the 2nd power

**cubed** – to the 3rd power

**PRACTICE**

*show your work • label units*

1. Find the value of each power.

**Step 1** Write each power in expanded form.

**Step 2** Multiply one factor at a time.

a.  $7^2 =$  \_\_\_\_\_

b.  $3^4 =$  \_\_\_\_\_

c.  $5^3 =$  \_\_\_\_\_

d.  $10^5 =$  \_\_\_\_\_

e.  $2^6 =$  \_\_\_\_\_

f.  $1^8 =$  \_\_\_\_\_

2. Find the value. Write fractions in simplest form.

a.  $(1/2)^3 =$  \_\_\_\_\_

b.  $0.3^2 =$  \_\_\_\_\_

3. Write each number as a power of the given base.

a. 27 as a power of 3: \_\_\_\_\_

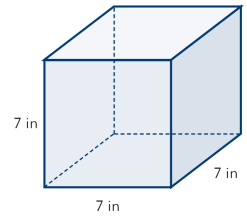
b. 10,000 as a power of 10: \_\_\_\_\_

4. **Storage Cube.** A storage cube has edges that are **7 inches** long. Its **volume** is  $7^3$  cubic inches. Find the volume.

**Step 1** Write  $7^3$  as  $7 \times 7 \times 7$ .

**Step 2** Multiply  $7 \times 7$  first, then multiply by 7.

Show your work.



**Answer** \_\_\_\_\_  $\text{in}^3$

5. **Compare.** Select the **two** powers that have the same value.

**Step 1** Find the value of each power.

**Step 2** Pick the two with the same value.

☐ **A**  $2^6$

☐ **B**  $6^2$

☐ **C**  $4^3$

☐ **D**  $3^4$

### TEST PRACTICE

choose the best answer

6. What is the **value** of  $4^3$ ? Write  $4 \times 4 \times 4$  first.

**A** 7

**B** 12

**C** 64

**D** 81

7. Which expression has a value of 1,000?

**A**  $10^3$

**B**  $3^{10}$

**C**  $10 \times 3$

**D**  $100^3$

#### CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #\_\_\_\_

**AT HOME** Ask someone at home to guess which is greater,  $2^5$  or  $5^2$ . Then find both values together.